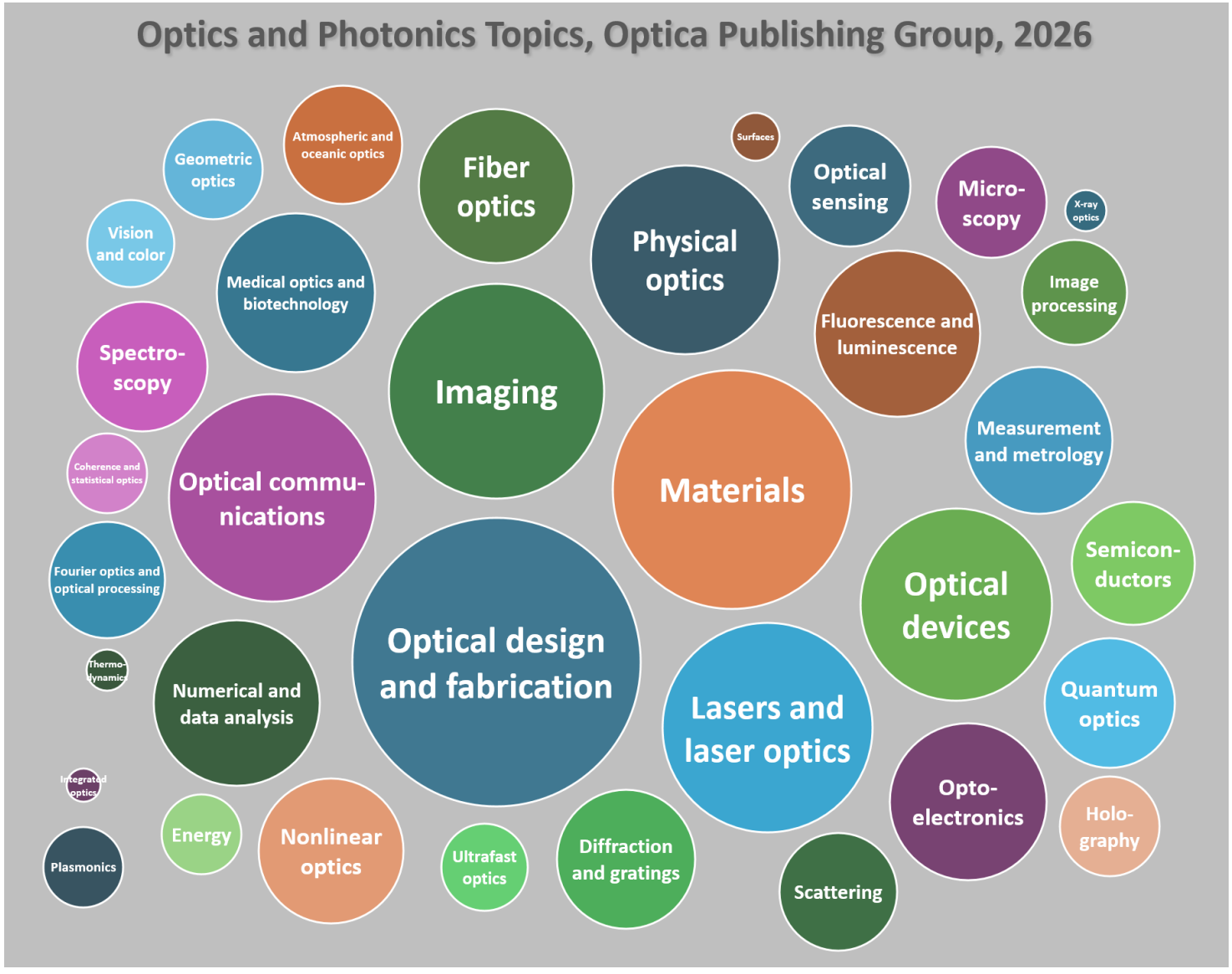
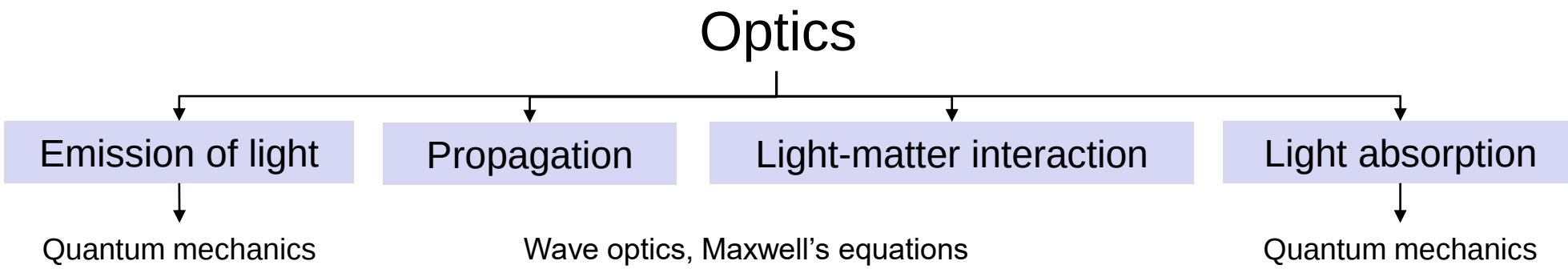
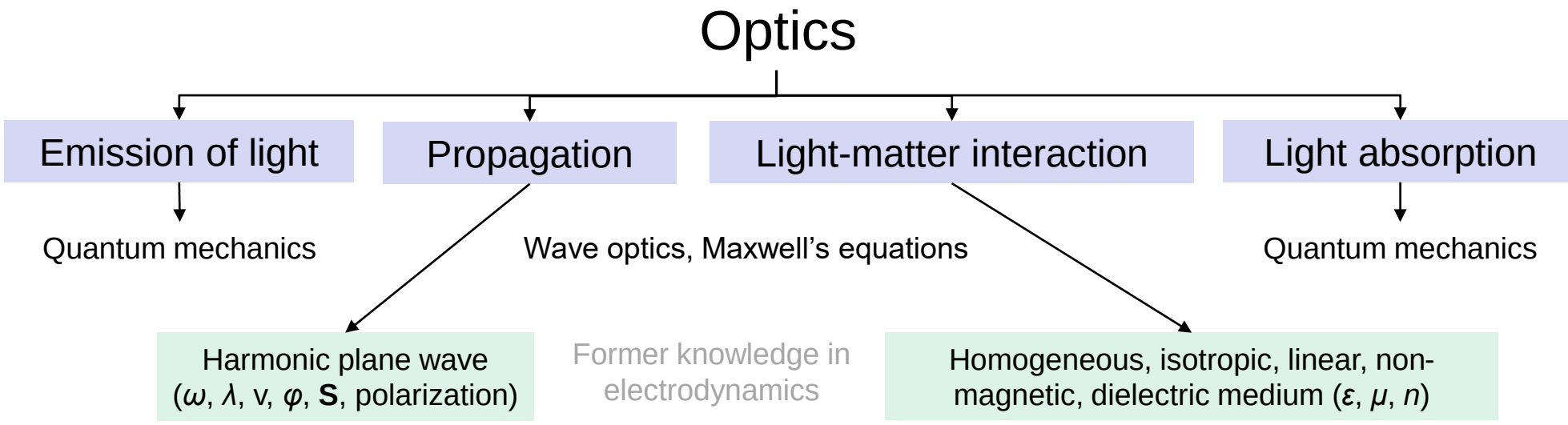
▪ Disciplines in optics ▪



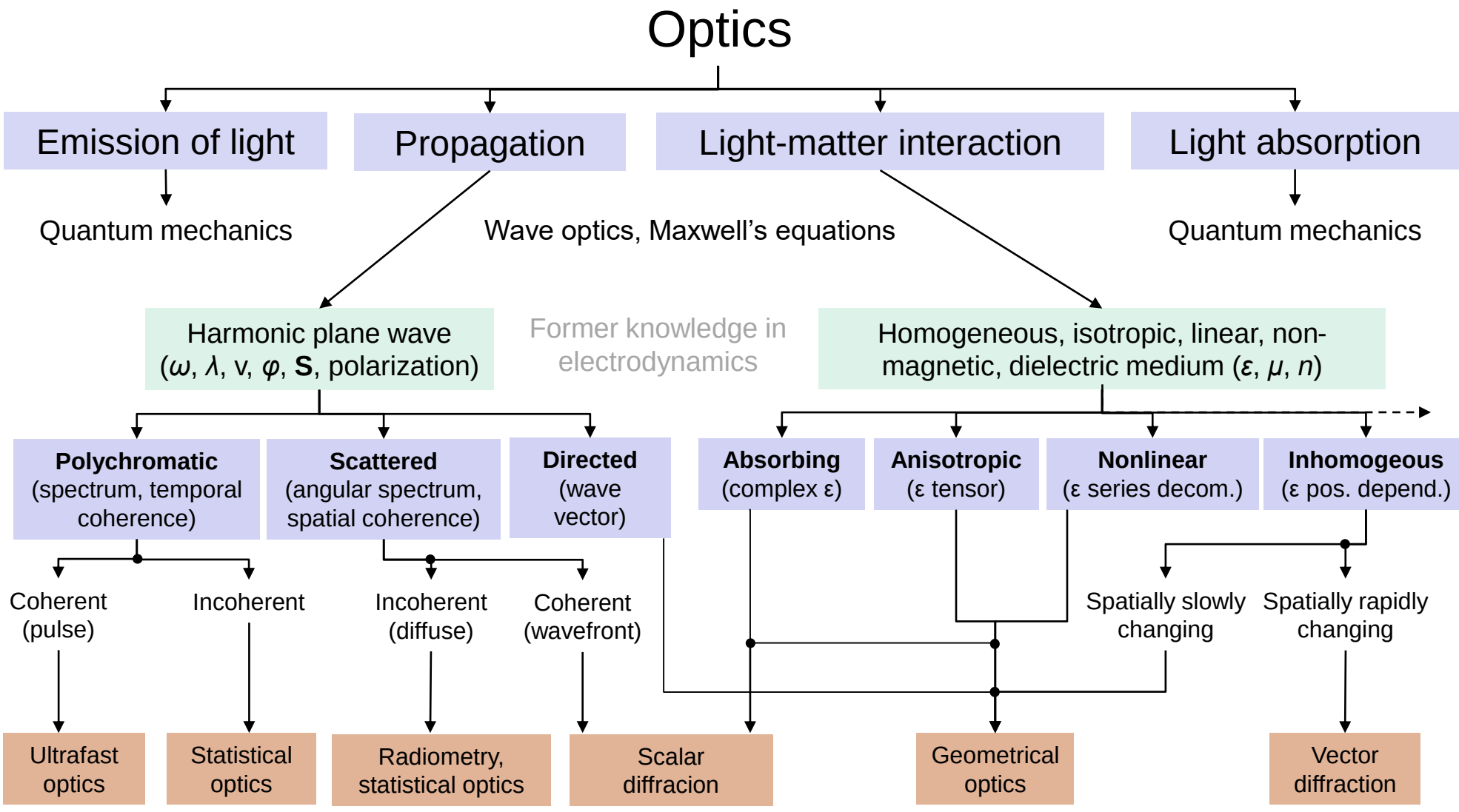
▪ Overviewing models in optics ▪



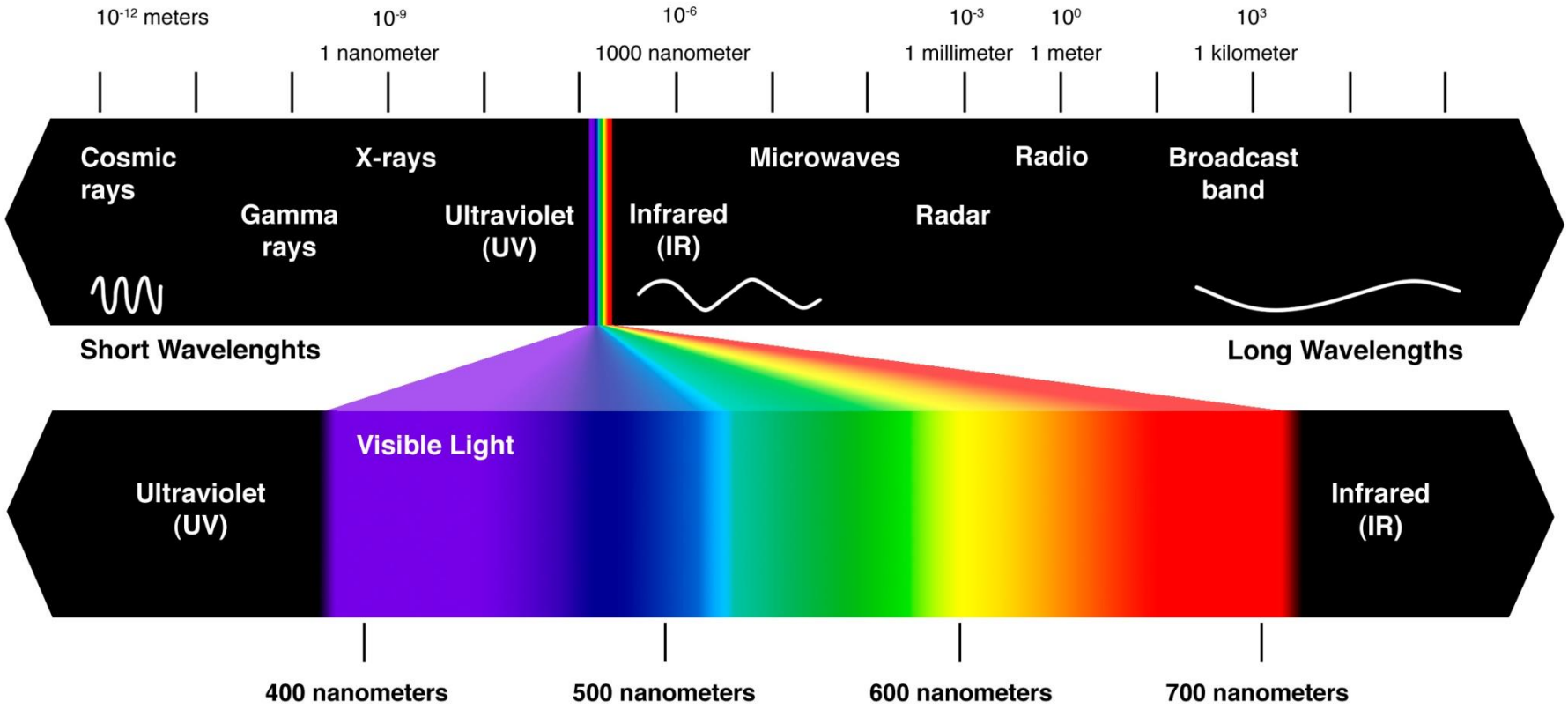
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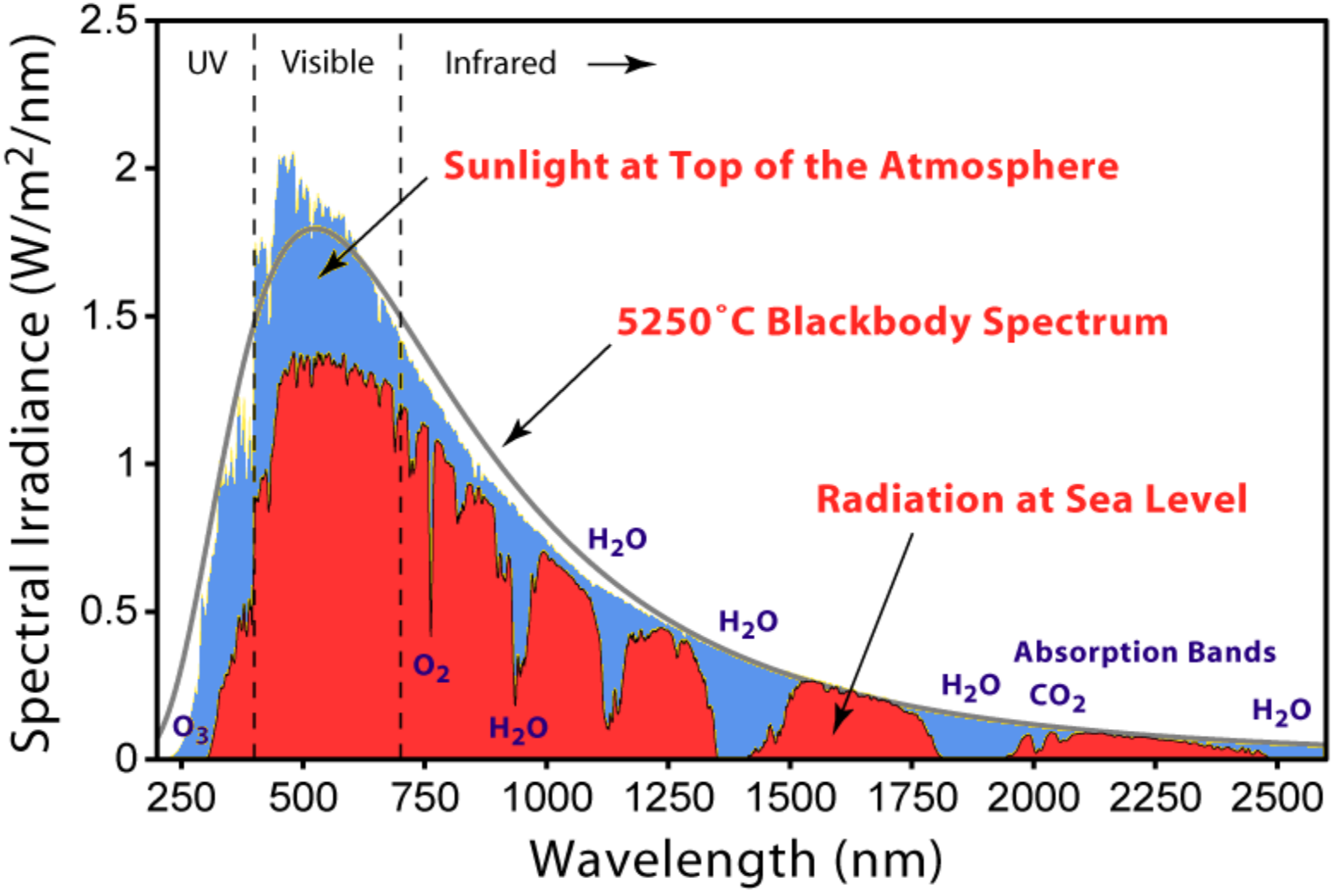


▪ Electromagnetic spectrum ▪



http://infothread.org/Science/Physics/Electromagnetic%20Spectrum.jpg

▪ Power spectral density of sunlight ▪

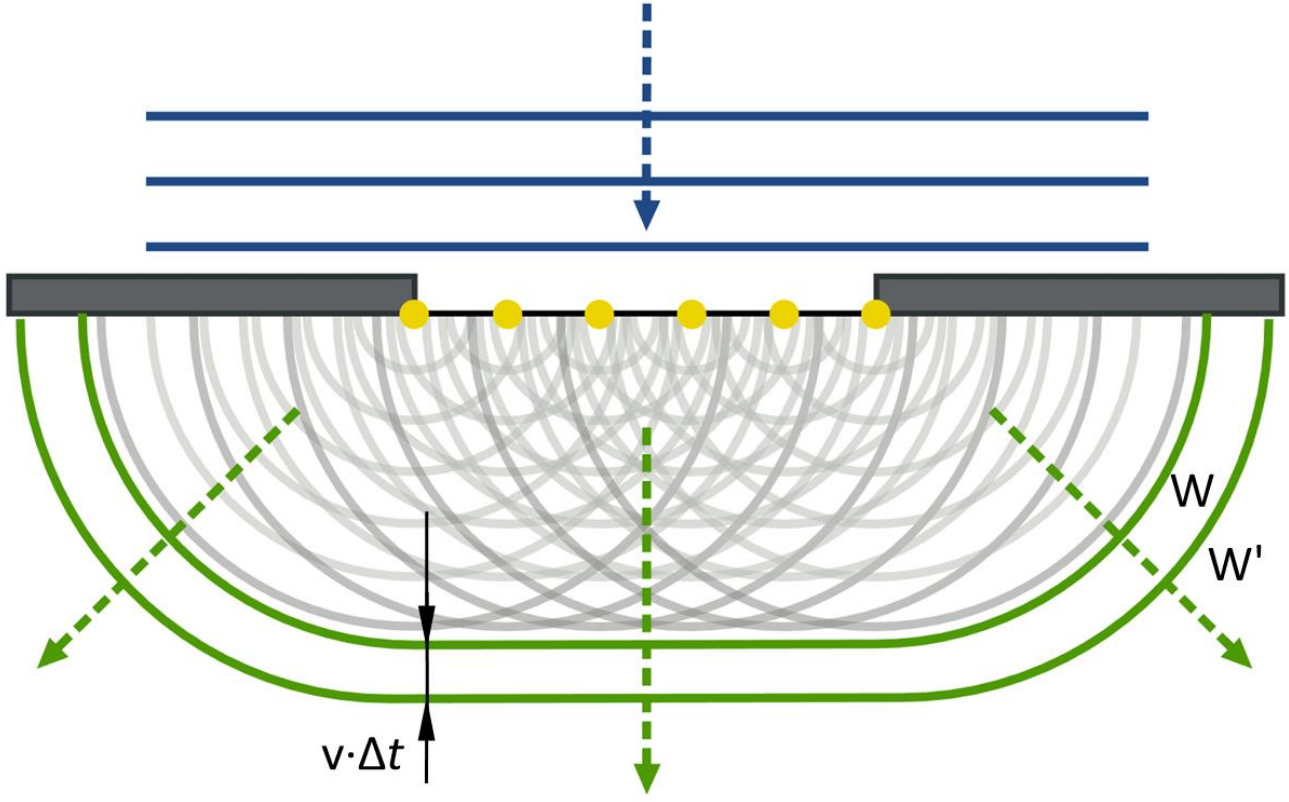


5500 K (5250°C) – direct sunlight on the surface, outside the atmosphere it is 5900 K
http://commons.wikimedia.org/wiki/File:Solar_Spectrum.png

▪ Discovery of light and EM waves ▪

Name	Age	Discovery
Euclid	280 BCE	“Optics”, light = vision (rays originate from the eye, propagate along straight lines
Alhazen	11 c.	light: originates from the source, eye is a “detector”; particle model; finite and density-dependent speed
Descartes	1637	elastic medium - aether, as if tennis balls make mechanical movement in it; law of recfraction
Fermat	1657	concept of the least time, a new derivation of law of refraction, light propagates slower in glass than in air
Rømer	1675	measuring light speed with 24% accuracy based on Jupiter moons (Doppler effect)
Huygens	1679	light is a disturbance propagating in an elastic medium, pulse
Newton	1704	“Opticks”; got the corpuscular model stuck for 100 years, light = sum of colors (coined “spectrum”)
Young	1801	declaring wave behaviour, color~wavelength, later: transversality, linear polarization
Fresnel	1818	diffraction experiments → mathematical description of the wave nature of light
Maxwell	1862	theory of electromagnetic waves (following Michael Faraday)
Michelson & Morley	1887	interferometric experiment → no absolute aether?
Hertz	1888	light is an EM wave – experimental proof (experiments with radio waves, speed, polarization)

▪ Huygens principle ▪

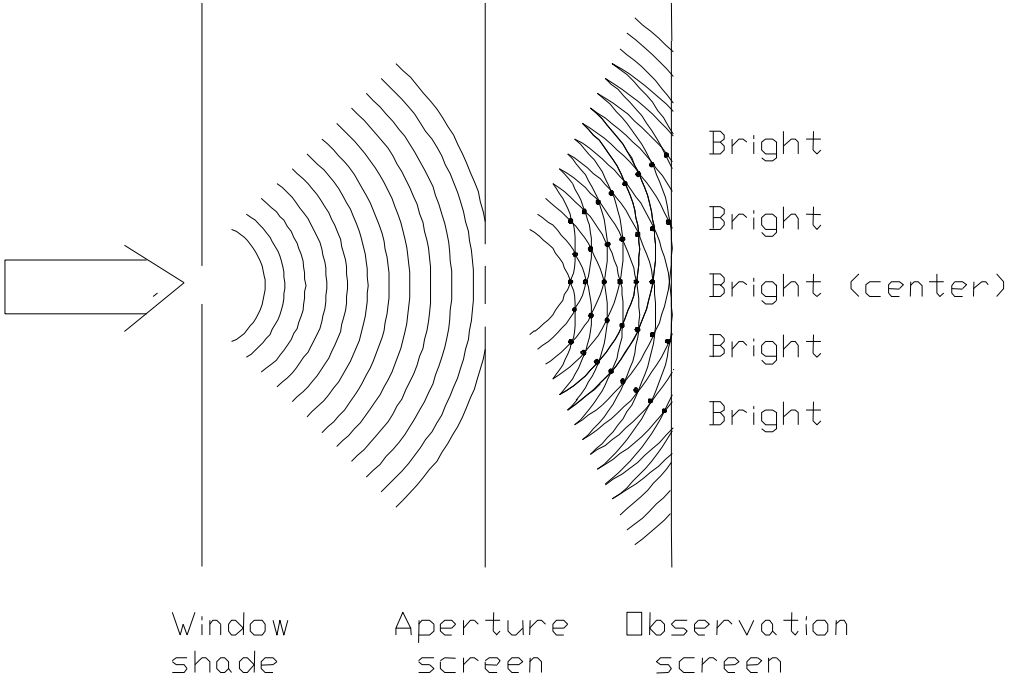


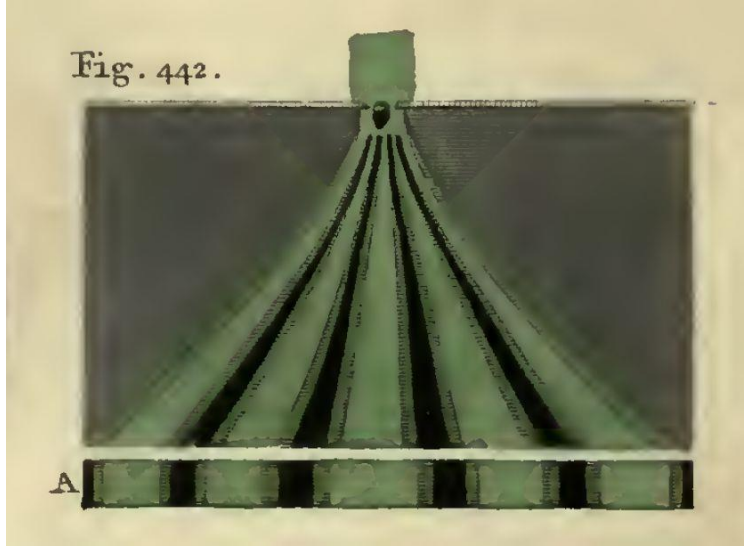
https://en.wikipedia.org/wiki/Huygens%E2%80%93Fresnel_principle

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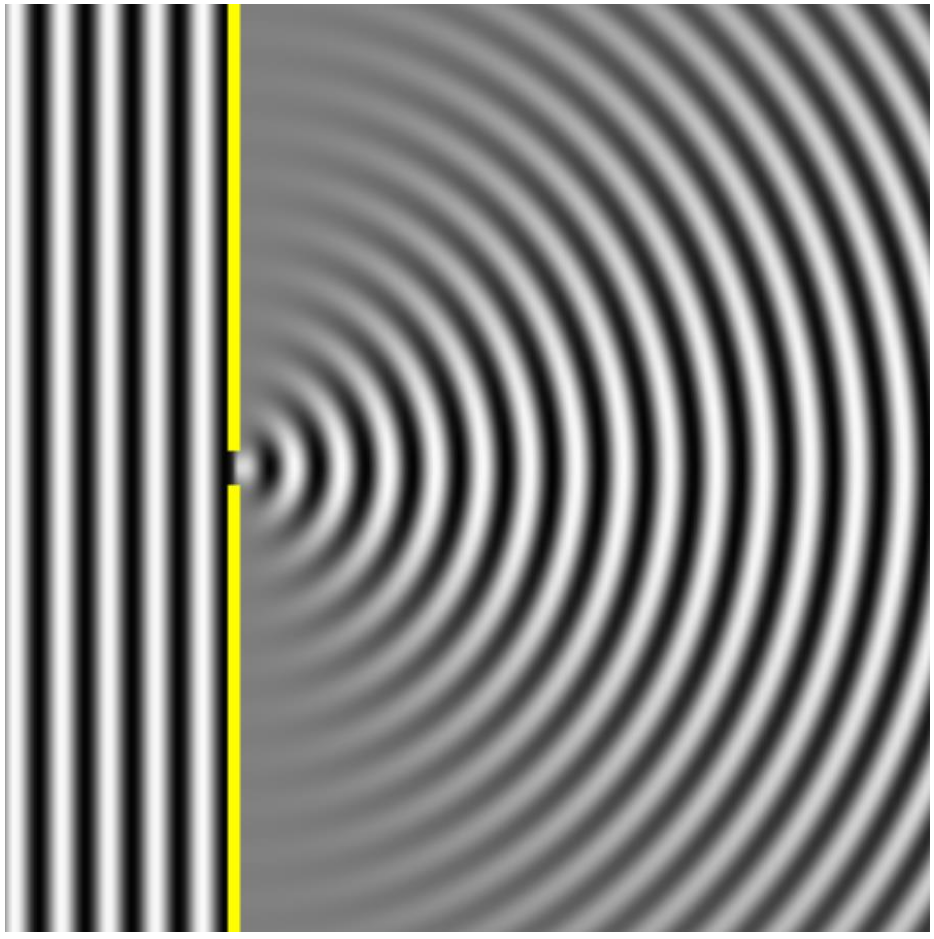
▪ Young's two-slit interference ▪





Forrás: T. Young, A Course of Lectures on Natural Philosphy and the Mechanical Arts, Vol. I, 1807

▪ Huygens-Fresnel principle → diffraction ▪

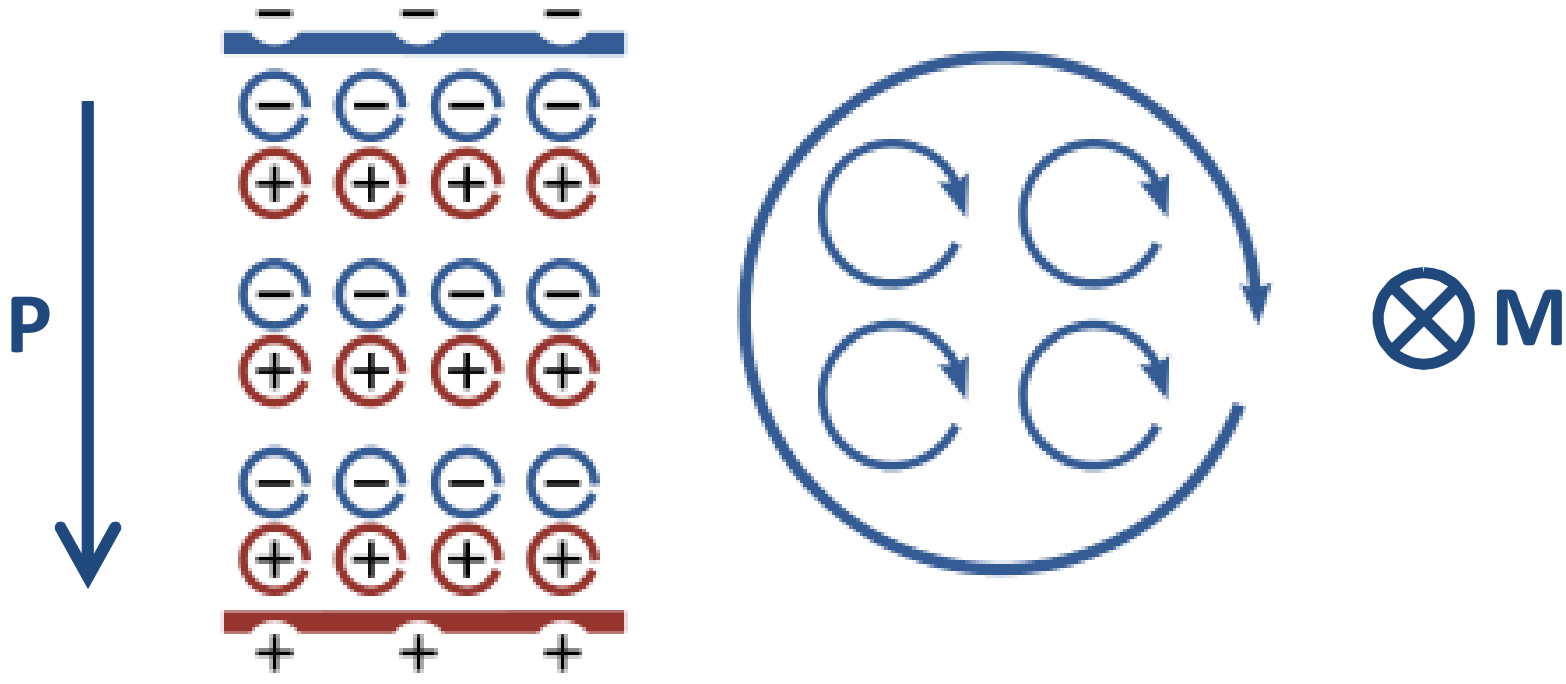


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▪ Discovery of light and EM waves ▪

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▪ Microscopic charges, currents ▪



Source: wikipedia

▪ Macroscopic Maxwell equations ▪

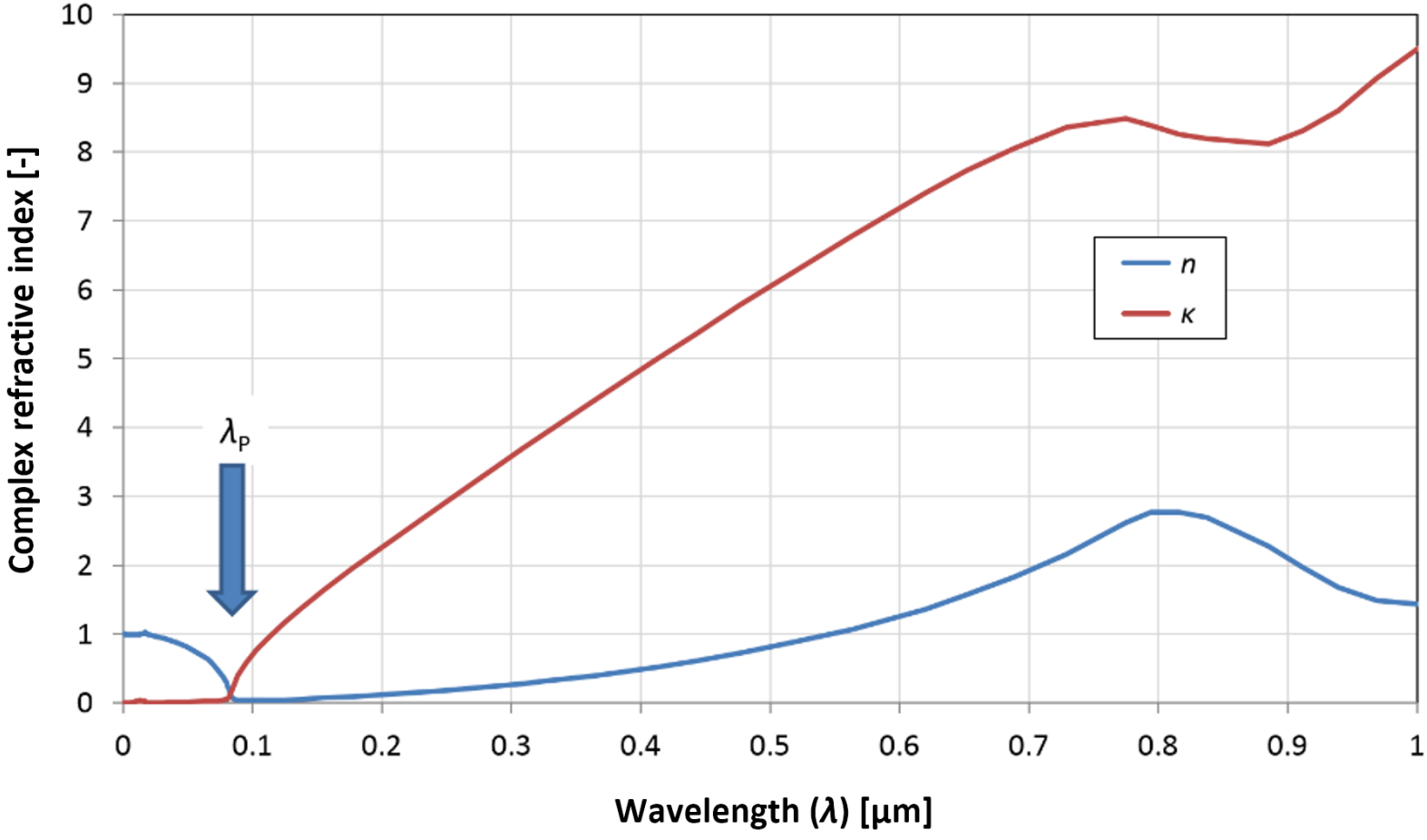
$$\left.\begin{aligned} 1)\ & \operatorname{curl}\mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{j} \\ 2)\ & \operatorname{curl}\mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ 3)\ & \operatorname{div}\mathbf{B} = 0 \\ 4)\ & \operatorname{div}\mathbf{D} = \rho \end{aligned}\right\}$$

$$\mathbf{B} = \mu_0 \mu_r \mathbf{H} = \mu \mathbf{H}$$

$$\mathbf{D} = \varepsilon_0 \varepsilon_r \mathbf{E} = \varepsilon \mathbf{E}$$

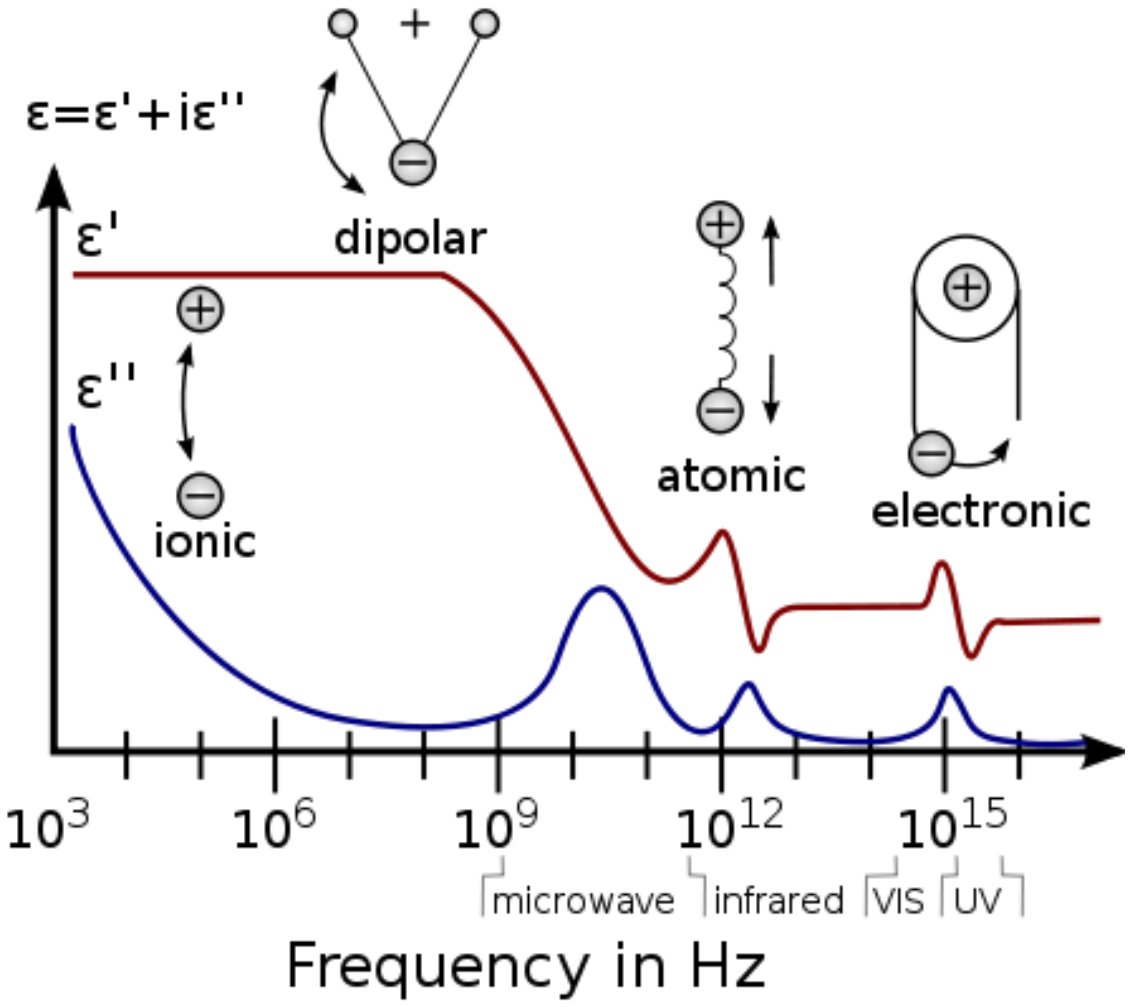
$$\mathbf{j} = \sigma \mathbf{E}$$

▪ Refractive index of aluminum ▪



(Source: http://refractiveindex.info)

▪ Frequency dependence of ε (fictive) ▪



(forrás: http://refractiveindex.info)